Nikhil N S

M.Tech Smart Manufacturing
CGPA: 8.2

ACADEMIC PROFILE

M Tech in Smart Manufacturing	8.2	Department of Design and Manufacturing	2026
B Tech	8.5	College of Agricultural Engineering	2024
Class XII(State)	84.16	M Basavaiah Residential Pu College, Sirigere	2020
Class X(Karnataka Secondary Education Examination Board)	82.56	Sri Ramalingeshwara swamy High School	2018

INTERNSHIPS — 2 Months

TCS Research Tata Consultancy Service Ltd. **Research Intern** **Jun'25-Aug'25**

- **Knowledge-guided optioneering for manufacturing process design**
- – Conducted a literature review on **Facility Layout Planning** (FLP) and generative design, reviewing 15+ papers.
  – Designed optimization models using **GA** in Pymoo for single- and multi-objective FLP.
  – Developed an optimization algorithm selection framework: **Neo4j KG, LangChain, RAG, Gemini LLM**.

PUBLICATIONS

- A **Knowledge Graph-based Retrieval-Augmented Generation** Framework for Algorithm Selection in the Facility Layout Problem. (*https://arxiv.org/abs/2509.18054*)

ACHIEVEMENTS

- Secured **AIR 13** in GATE 2024 (**Score: 808**)
- Achieved **3rd place** for Smart Irrigation Project in institute-level competition among **28 teams**, & was also selected for **Smart India Hackathon.**

PROJECTS

AI-Driven Application for High-Temperature Die-Casting (Nationally funded)-MTech **Sep'25-**

- Designed a **small-data** ML pipeline for squeeze-casting optimization under limited data using **LHS sampling, LASSO** as ML model, with **BO**, and achieved **R^2 ≈ 0.82 (UTS) & 0.845 (EL)** on simulated data using **scikit-learn, NumPy, Pandas, BoTorch**.
- **Ongoing:** Designing an end-to-end **Digital Twin** architecture to predict and mitigate casting defects, from high-frequency IoT sensor data via real-time, **physics-informed ML** models.

Sentiment Analysis of Text using Generative & Discriminative Models **Mar'25-Mar'25**

- Implemented **Naive Bayes classifier** from scratch with custom text preprocessing, **BoW** feature extraction, & Laplace smoothing.
- Trained a **Logistic Regression** classifier using **BoW and Word2Vec** embeddings, achieving **≈80%** validation accuracy.
- Built a **semantic analysis** leveraging pre-trained Word2Vec to compute word analogies & similarity using **cosine & Euclidean dist**.

Data Analysis on Web Scraped Smartprix Mobile Dataset **Aug'25-Sep'25**

- Collected and structured smartphone **data from Smartpix** using a custom **Selenium–BeautifulSoup scraper**, enabling large-scale analysis of mobile brand and feature trends.
- Performed extensive **data cleaning, EDA, feature engineering, construction, and transformation** to uncover key trends and correlations influencing smartphone pricing.
- Developed and compared multiple ML models, including a **Stacking Regressor** using scikit-learn and achieved a high **R^2≈ 0.92** in smartphone price prediction. (*https://github.com/NikhilNS26/Data-analysis-with-web-scrapped-data*)

Edge AI–based Precision Irrigation System **Mar'25-May'25**

- Processed **34 yrs of climate data** with Z-score normalization & **correlation analysis** to identify key predictors of crop water demand.
- Trained a Multi-Layer Perceptron (**MLP**) regressor in scikit-learn to estimate daily reference evapotranspiration (**ET**), achieving **R^2 ≈ 0.97**. Deployed a lightweight model on Raspberry Pi Pico.(*https://github.com/NikhilNS26/Smart-irrigation-system*)

MISCELLANEOUS

- **Academic Assignments:**Analyzed air quality data (2017–2020) for Delhi & Bangalore using **time-series & correlation analysis, regression, and PCA** to evaluate COVID-19 lockdown impacts on AQI.
- Validated the **Central Limit Theorem** through simulation and performed statistical **hypothesis testing** with SciPy.(*https://github.com/NikhilNS26/Data-Science-Case_studies/tree/main*)

SKILLS

- **Programming Languages:** Python
- **Frameworks & Tools:** Langchain, Numpy, Pandas, Pymoo, PyTorch, Tensorflow/keras, Scikit-learn, Seaborn.
- **Technical:** Machine Learning, Deep Learning, GenAI, Transformer.

COURSES

(1) Data Analysis, (2) Data mining, (3) Machine Learning, (4) Edge AI, (PG)
(5) Introduction to Natural Language Processing, (6) Intelligent Mobile Robots, (7) Digital Manufacturing, (PG)